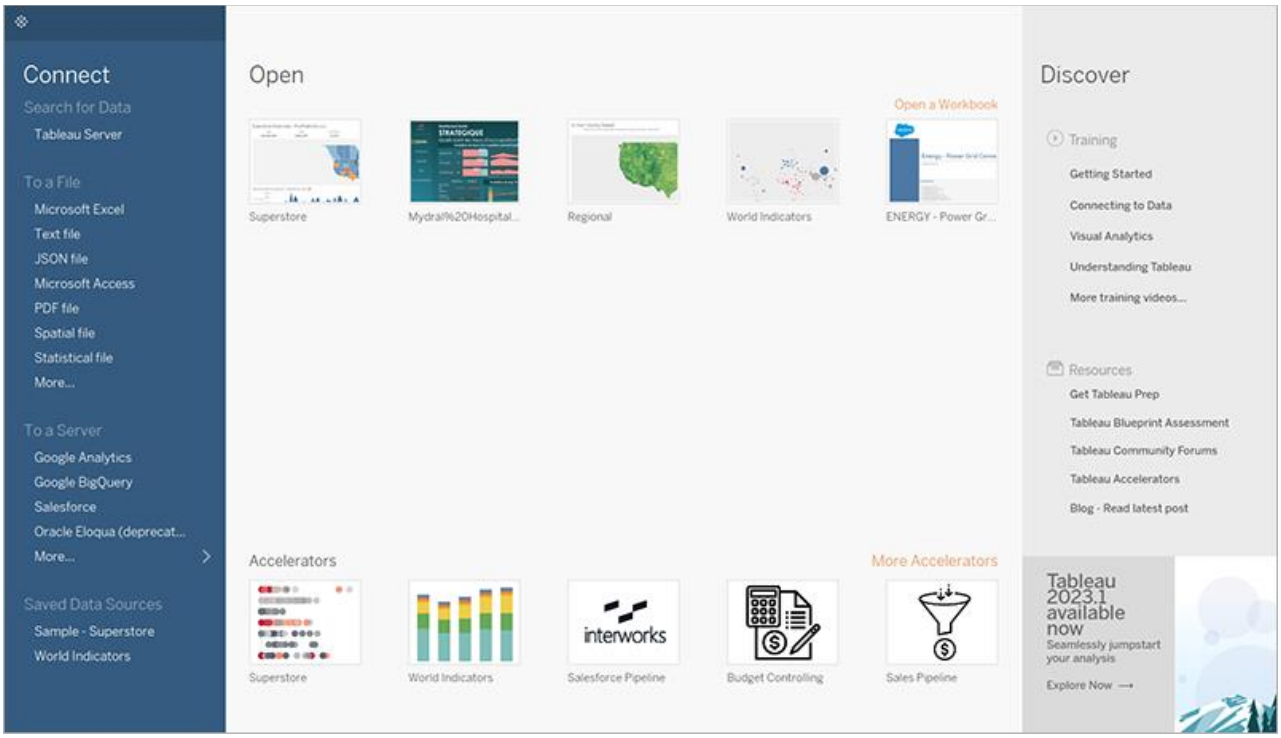
PCA25D06J Data Visualization Techniques.

Exercise 1: Installing Tableau

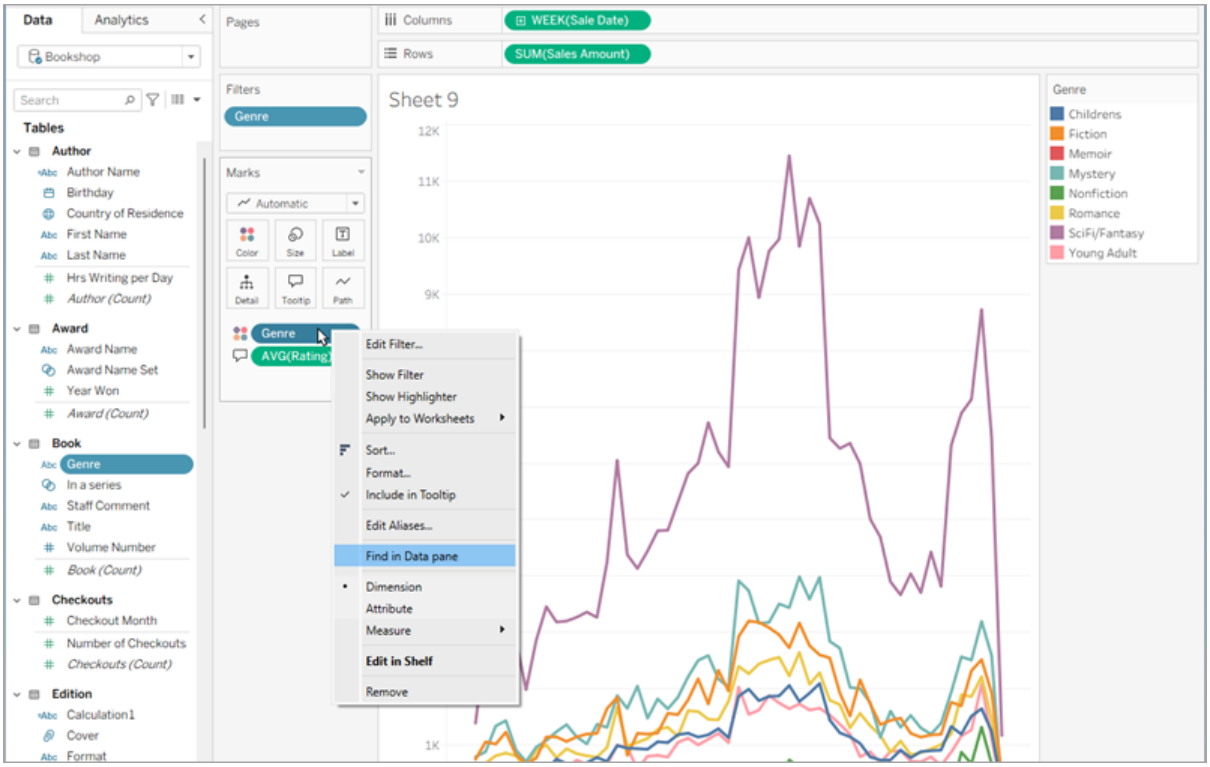
- **Aim:** To successfully download and install Tableau Public / Tableau Desktop on the local machine.
- **Procedure:**
  1. Navigate to the official Tableau website (public.tableau.com).
  2. Enter your email address and click on "Download the App".
  3. Locate the downloaded installer file and run it.
  4. Follow the on-screen installation prompts and accept the license agreement.
  5. Launch the Tableau application once the installation is complete.
- **Code:** Not Applicable (GUI-based installation).



- **Output:** The Tableau Welcome Screen appears, displaying the "Connect" pane.
- **Result:** Tableau was successfully installed and launched on the system.

Exercise 2: Working with a sample dataset in a Tableau Workspace

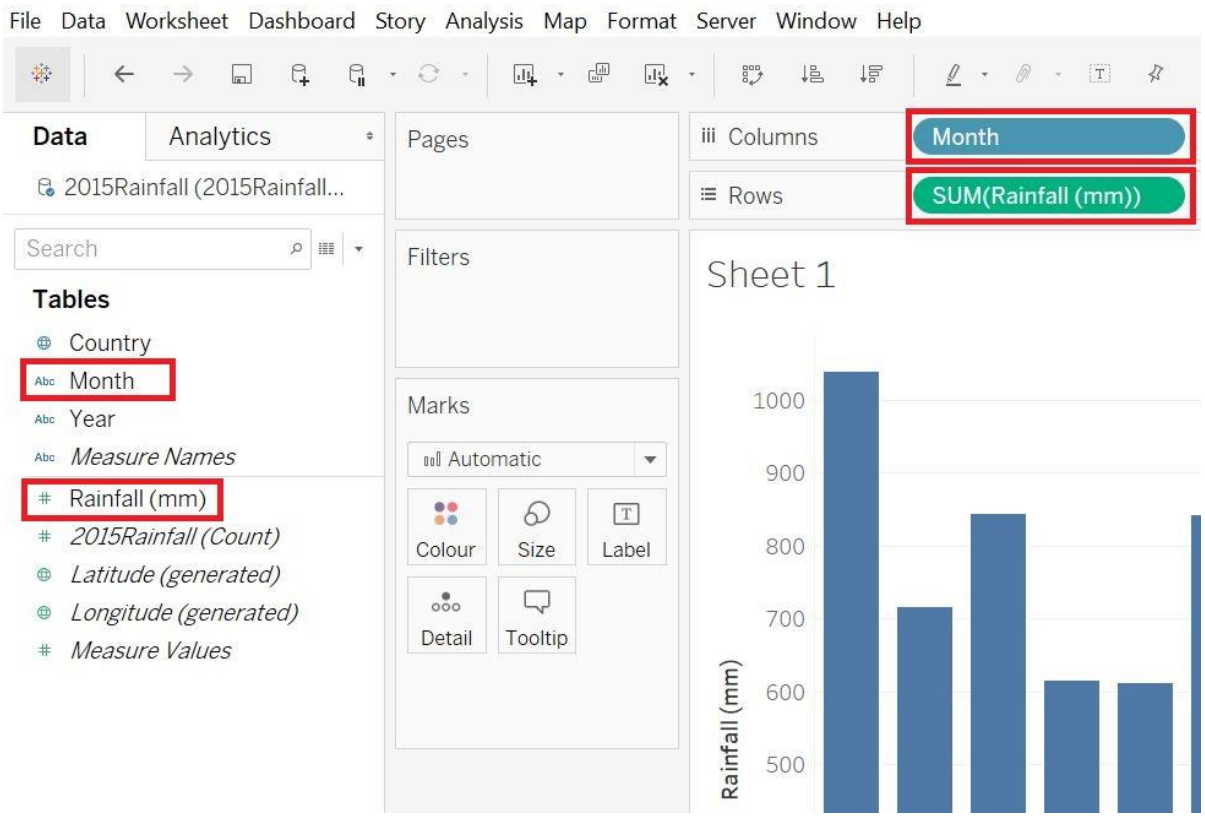
- **Aim:** To connect Tableau to a sample dataset and familiarize oneself with the Tableau workspace.
- **Procedure:**
  1. Open Tableau and under the "Connect" pane, click on "Microsoft Excel" or "Text file" depending on the dataset.
  2. Browse and select the "Sample - Superstore" dataset.
  3. Drag the "Orders" sheet to the canvas to view the data preview.
  4. Click on "Sheet 1" at the bottom to enter the main Tableau workspace.
- **Code:** Not Applicable (GUI-based operation).



- **Output:** The workspace opens, displaying dimensions (categorical data) and measures (numerical data) on the left data pane.
- **Result:** The sample dataset was successfully loaded and is ready for visualization.

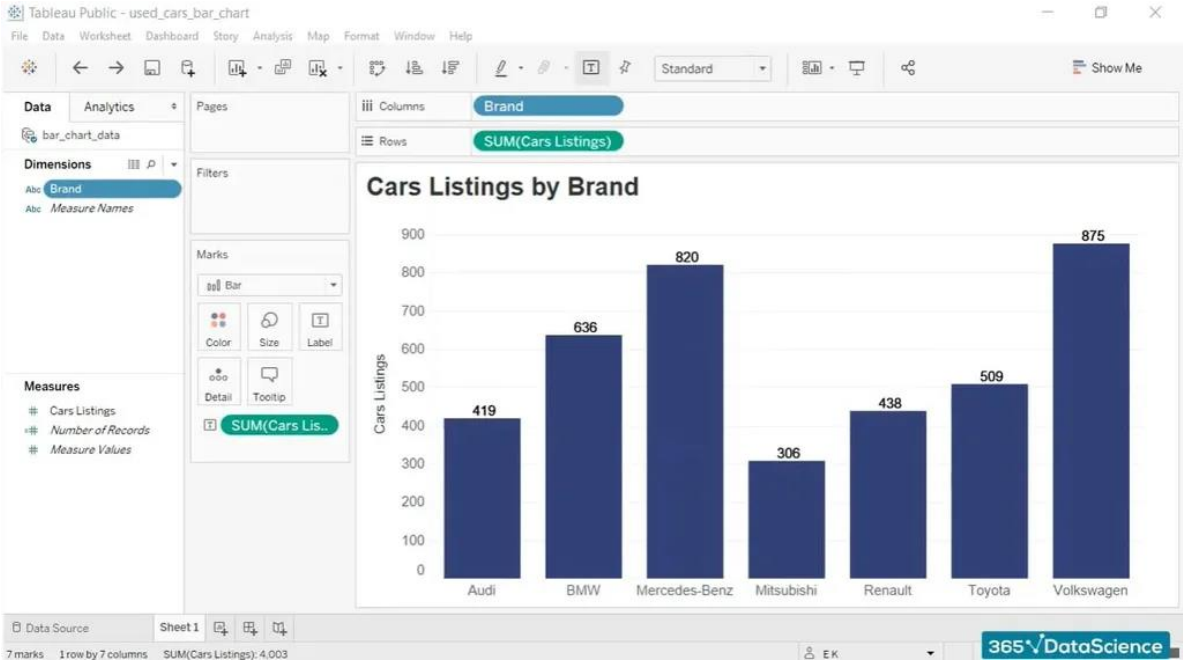
Exercise 3: Working with Data Tables

- **Aim:** To create and format a basic text table (cross-tab) to summarize data.
- **Procedure:**
  1. In Sheet 1, drag a dimension (e.g., Category) to the **Rows** shelf.
  2. Drag another dimension (e.g., Region) to the **Columns** shelf.
  3. Drag a measure (e.g., Sales) to the **Text** icon on the Marks card.
  4. Format the table by adding grand totals via the Analytics pane.
- **Code:** Not Applicable (Drag-and-drop operation).



- **Output:** A text table displaying total sales categorized by product category and region.
- **Result:** A data table was successfully created and formatted.

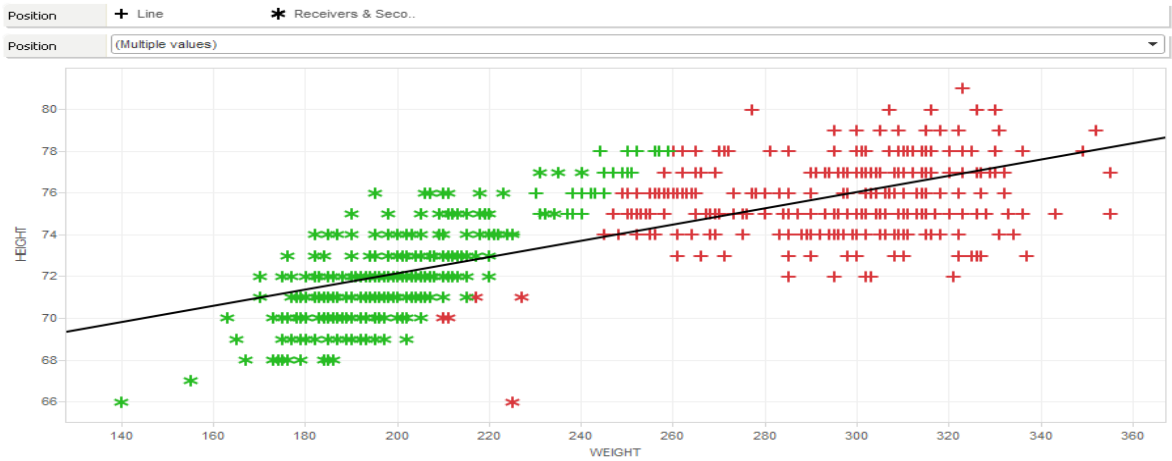
Exercise 4: Graphical tools for data elaboration - 1

- **Aim:** To elaborate data using fundamental graphical tools like Bar Charts and Line Charts.
- **Procedure:**
    1. **Bar Chart:** Drag Sub-Category to Rows and Sales to Columns. Sort the chart in descending order.
    2. Open a new worksheet.
    3. **Line Chart:** Drag Order Date to Columns (expand to Month) and Profit to Rows to see the trend over time.
- **Code:** Not Applicable.
- 

- **Output:** Two visualizations: A sorted bar chart comparing sales by sub-category, and a line chart showing profit trends over time.
- **Result:** Basic data elaboration charts were successfully generated.

Exercise 5: Graphical tools for data elaboration - 2

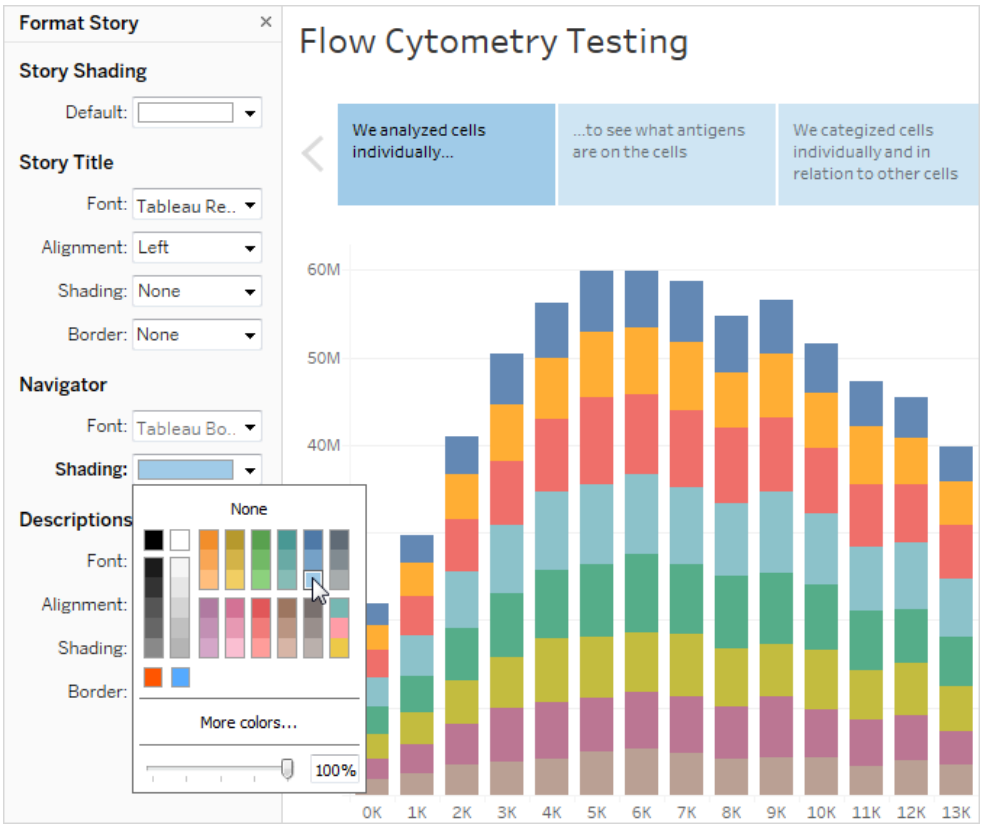
- **Aim:** To elaborate data using advanced graphical tools like Scatter Plots and Heat Maps.
- **Procedure:**
    1. **Scatter Plot:** Drag Sales to Columns and Profit to Rows. Drag Customer Name to the Detail mark to create a scatter plot.
    2. Open a new worksheet.
    3. **Heat Map:** Drag Region to Columns and Category to Rows. Drag Profit to the Color mark and change the mark type to Square.
- **Code:** Not Applicable.



- **Output:** A scatter plot identifying customer profitability, and a heat map showing the density of profit across regions and categories.
- **Result:** Advanced data elaboration charts were successfully generated.

Exercise 6: Create a story with Tableau

- **Aim:** To create a data story that presents a narrative using a sequence of visualizations.
- **Procedure:**
  1. Click the "New Story" button at the bottom of the workspace.
  2. Drag previously created worksheets (e.g., Bar chart, Line chart) onto the story canvas.
  3. Add a caption to the first story point (e.g., "Sales by Category").
  4. Click "Blank" to add a new story point, drag the next worksheet, and add a caption.
  5. Adjust the layout and text to complete the narrative.
- **Code:** Not Applicable.



- **Output:** A multi-point interactive presentation layout inside Tableau.
- **Result:** A data story was successfully created to guide users through the insights.

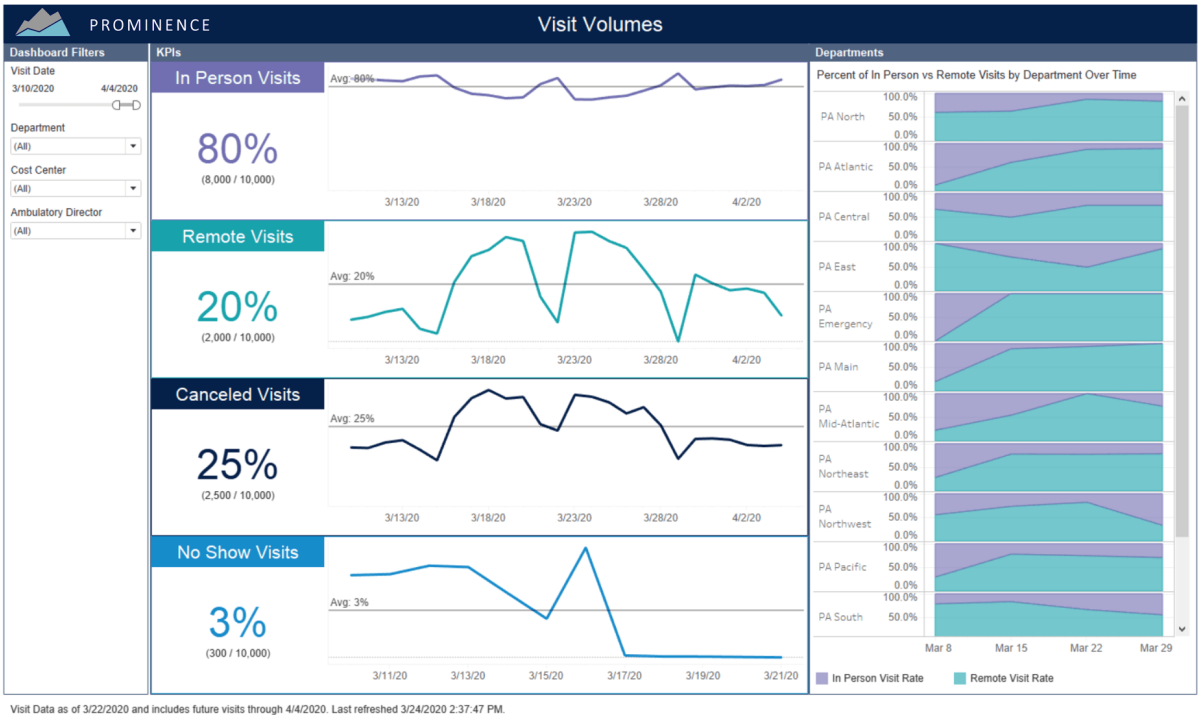
Exercise 7: Creating a new dashboard

- **Aim:** To combine multiple worksheets into a single, cohesive dashboard.
- **Procedure:**
  1. Click the "New Dashboard" icon at the bottom.
  2. From the left pane, drag and drop multiple created worksheets (e.g., Map, Bar Chart, Line Chart) onto the blank dashboard canvas.
  3. Adjust the size of the dashboard to "Automatic" or a specific resolution.
  4. Arrange the layout using layout containers (horizontal/vertical).
- **Code:** Not Applicable.

- **Output:** A single dashboard displaying multiple charts simultaneously.
- **Result:** A new dashboard was successfully created.

Exercise 8: Working with a Dashboard

- **Aim:** To add interactivity to a dashboard using filters and actions.
- **Procedure:**
  1. On the created dashboard, click on one of the charts (e.g., a Map).
  2. Click the funnel icon ("Use as Filter") on the top right of the chart frame.
  3. Click on different states on the map to see the other charts update dynamically.
  4. Go to Dashboard > Actions to add highlight or URL actions.
- **Code:** Not Applicable.



- **Output:** An interactive dashboard where selecting data on one chart filters the data on the others.
- **Result:** Dashboard interactivity was successfully implemented.

Exercise 9: Building a real-time dashboard

- **Aim:** To configure a dashboard to reflect live or frequently updated data.
- **Procedure:**
    1. When connecting to the data source (e.g., an SQL database), choose the "Live" connection option instead of "Extract".
    2. Build the necessary worksheets and assemble them into a dashboard.
    3. Publish the dashboard to Tableau Server or Tableau Cloud.
    4. Configure the data refresh schedules or rely on the live query execution.
- **Code:** Not Applicable.
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- **Output:** A dashboard published to the server that automatically updates as the backend database receives new data.
- **Result:** A real-time data dashboard was successfully established.

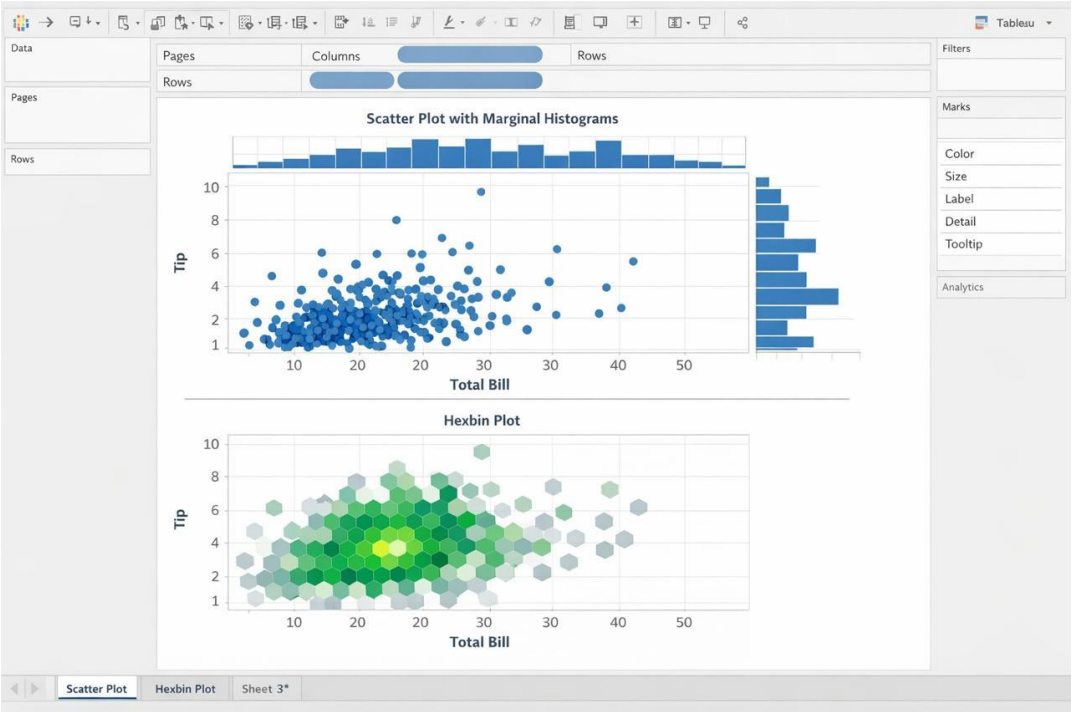
Exercise 10: Explore different types of Bivariate distributions

(Note: As the syllabus later introduces Python libraries like Seaborn and Matplotlib for statistical plots, this exercise utilizes Python code for bivariate distributions).

- **Aim:** To analyse the relationship between two variables by plotting bivariate distributions using Python and Seaborn.
- **Procedure:**
    1. Open a Jupyter Notebook or Python IDE.
    2. Import the seaborn and matplotlib libraries.
    3. Load a built-in dataset (e.g., 'tips').
    4. Use sns.jointplot() to create a scatter plot with marginal histograms.

Code:

```
import seaborn as sns
import matplotlib.pyplot as plt
# Load sample dataset
df = sns.load_dataset('tips')
# Plot Bivariate distribution (Scatter with histograms)
sns.jointplot(x='total_bill', y='tip', data=df, kind='scatter', color='blue')
plt.show()
# Plot Bivariate distribution (Hexbin)
sns.jointplot(x='total_bill', y='tip', data=df, kind='hex', color='green')
plt.show()
```



- **Output:** Two plots generated on the screen: one showing a scatter layout with marginal distributions on the axes, and another showing a hexagonal binning plot representing data density.
- **Result:** Different types of bivariate distributions were successfully explored and plotted using Python.